Lab 1 - SVM
From Margin to Mini-Batch Updates

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Lab Overview & Attendance

Part 1: Lab Session

- **Briefing:** Quick introduction to the lab and related concepts.
- **Hands-on Coding:** Complete the lab using the provided template code.
  - You are allowed to use LLMs.
  - Feel free to discuss with your neighbours or ask me questions.
- **Sample Solutions:** I will walk through the sample solutions and results.

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Part 2: Quiz

- **1-Hour Quiz** (**5%** of the final mark) using Examplify with the Internet blocked.
- **NO LLMs allowed** (including local ones).
- You can refer to all your local materials (e.g., slides, code, notes).

1.1 Binary SVM Geometry

For each training sample, $\boldsymbol{x}_i \in \mathbb{R}^d, y_i \in \{-1, +1\}$, the SVM uses three parallel hyperplanes:

- **Decision boundary**

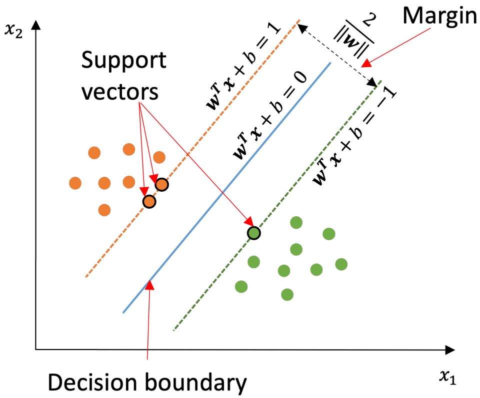
$$H_0 : \ \boldsymbol{w}^T \boldsymbol{x} + b = 0.$$

- **Positive supporting hyperplane**

$$H_+ : \ \boldsymbol{w}^T \boldsymbol{x} + b = +1.$$

- **Negative supporting hyperplane**

$$H_- : \ \boldsymbol{w}^T \boldsymbol{x} + b = -1.$$



1.2 The Half-Margin Is $1/\|\boldsymbol{w}\|$

Choose $\boldsymbol{x}_+ \in H_+$ and its projection $\boldsymbol{x}_0 \in H_0$. Their distance d follows the unit normal vector:

$$\boldsymbol{x}_+ - \boldsymbol{x}_0 = d\,\frac{\boldsymbol{w}}{\|\boldsymbol{w}\|}.$$

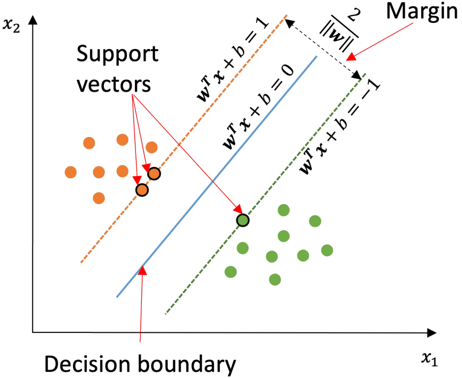
Using the equations of H_+ and H_0,

$$\boldsymbol{w}^T\boldsymbol{x}_+ + b = 1,$$
$$\boldsymbol{w}^T\boldsymbol{x}_0 + b = 0,$$
$$\boldsymbol{w}^T(\boldsymbol{x}_+ - \boldsymbol{x}_0) = 1.$$

Substitute the displacement:

$$1 = d\,\frac{\boldsymbol{w}^T\boldsymbol{w}}{\|\boldsymbol{w}\|} = d\,\|\boldsymbol{w}\|,$$

$$\boxed{d = \frac{1}{\|\boldsymbol{w}\|}}.$$



1.3 Maximizing the Margin Gives a Quadratic Objective

The distance from H_0 to either supporting hyperplane is

$$\frac{1}{\|\boldsymbol{w}\|}.$$

Hence, the full margin between H_+ and H_- is

$$\boxed{\text{margin} = \frac{2}{\|\boldsymbol{w}\|}}.$$

Maximizing this margin is equivalent to minimizing $\|\boldsymbol{w}\|$. Together with the separation constraints introduced next, we use the objective

$$\boxed{\min_{\boldsymbol{w},b} \ \frac{1}{2}\|\boldsymbol{w}\|^2}.$$

The factor $1/2$ does not change the minimizer; it makes the gradient simply $\boldsymbol{w}$.

2.1 Hard-Margin SVM for Separable Data

For a positive sample, we require

$$\boldsymbol{w}^T\boldsymbol{x}_i + b \geq 1 \qquad (y_i = +1).$$

For a negative sample, we require

$$\boldsymbol{w}^T\boldsymbol{x}_i + b \leq -1 \qquad (y_i = -1).$$

Then we can combine both cases into one constraint:

$$y_i\left(\boldsymbol{w}^T\boldsymbol{x}_i + b\right) \geq 1.$$

Thus, for linearly separable data, the hard-margin SVM is

$$\boxed{\begin{array}{ll} \min\limits_{\boldsymbol{w},b} & \dfrac{1}{2}\|\boldsymbol{w}\|^2 \\ \text{subject to} & y_i\left(\boldsymbol{w}^T\boldsymbol{x}_i + b\right) \geq 1, \quad i = 1, \ldots, N. \end{array}}$$

2.2 Slack Variables Permit Margin Violations

If the data are not perfectly separable, the hard-margin constraints may have no feasible solution. Thus, we introduce one slack/buffer variable $\xi_i \geq 0$ per sample.

- For $y_i = +1$:

$$\boldsymbol{w}^T \boldsymbol{x}_i + b \geq 1 - \xi_i.$$

- For $y_i = -1$:

$$\boldsymbol{w}^T \boldsymbol{x}_i + b \leq -1 + \xi_i.$$

Similarly, we can combine both cases:

$$\boxed{y_i \left(\boldsymbol{w}^T \boldsymbol{x}_i + b \right) \geq 1 - \xi_i, \qquad \xi_i \geq 0.}$$

Larger ξ_i means a larger **violation** of the desired margin constraint.

2.3 Soft-Margin SVM Balances Width and Violations

The soft-margin SVM penalizes slack while still preferring a wide margin:

$$\boxed{\begin{array}{ll} \displaystyle\min_{\boldsymbol{w},b,\boldsymbol{\xi}} & \displaystyle\frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{N}\xi_i \\ \text{subject to} & y_i\left(\boldsymbol{w}^T\boldsymbol{x}_i + b\right) \geq 1 - \xi_i, \\ & \xi_i \geq 0, \qquad i = 1, \ldots, N. \end{array}}$$

- $\frac{1}{2}\|\boldsymbol{w}\|^2$ favors a wider margin.
- $\sum_i \xi_i$ penalizes margin violations.
- $C > 0$ regularization parameter C controls the trade-off between these two goals.

2.4 Let's Look at the Constraints (Hinge Loss)

Rearranging the margin constraint gives

$$\begin{cases} \xi_i \geq 1 - y_i\left(\boldsymbol{w}^T \boldsymbol{x}_i + b\right) \\ \xi_i \geq 0 \end{cases}$$

And based on the objective func., we want to minimize ξ_i, thus the minimum possible ξ_i is the larger of the 2 lower boundaries:

$$\boxed{\xi_i = max\left\{0, 1 - y_i\left(\boldsymbol{w}^T \boldsymbol{x}_i + b\right)\right\}.}$$

Finally, put the ξ_i back to the objective func., and can get the unconstrained objective

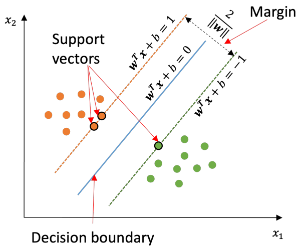
$$\boxed{J(\boldsymbol{w}, b) = \underbrace{\frac{1}{2}\|\boldsymbol{w}\|^2}_{\text{wider margin}} + C \sum_{i=1}^{N} \underbrace{\max\left\{0, 1 - y_i\left(\boldsymbol{w}^T \boldsymbol{x}_i + b\right)\right\}}_{\substack{\ell_i\text{: hinge loss:} \\ \text{measure how much a sample violates the margin}}} .}$$

2.5 Understanding the Hinge Loss

For sample i, the hinge loss is

$$\ell_i = \max\left\{0, 1 - y_i\left(\boldsymbol{w}^T \boldsymbol{x}_i + b\right)\right\}.$$

Value of $y_i(\boldsymbol{w}^T\boldsymbol{x}_i + b)$	Interpretation
≥ 1	Correctly classified; on or beyond the supporting hyperplane; $\ell_i = 0$
$0 < \cdot < 1$	Correctly classified; but inside/violate the margin; $0 < \ell_i < 1$
$= 0$	On the decision boundary; $\ell_i = 1$
< 0	Misclassified; $\ell_i > 1$



Hinge loss penalizes not only wrong predictions, but also correct predictions that are too close to the decision boundary.

3.1 Two Cases of the Single-Sample Objective

For one sample $(\boldsymbol{x}_i, y_i)$, use

$$L_i(\boldsymbol{w}, b) = \frac{1}{2}\|\boldsymbol{w}\|^2 + C \max\Big\{0, 1 - y_i\big(\boldsymbol{w}^T\boldsymbol{x}_i + b\big)\Big\}.$$

The subgradient depends on whether the sample violates the unit-margin constraint:

1. **No violation:** $y_i(\boldsymbol{w}^T\boldsymbol{x}_i + b) \geq 1.$ The hinge term is zero.
2. **Violation:** $y_i(\boldsymbol{w}^T\boldsymbol{x}_i + b) < 1.$ The hinge term is active.

3.2 Case 1: No Violation Gradient and Update Rule

If

$$y_i\left(\boldsymbol{w}^T \boldsymbol{x}_i + b\right) \geq 1,$$

then

$$L_i(\boldsymbol{w}, b) = \frac{1}{2}\|\boldsymbol{w}\|^2.$$

Therefore,

$$\nabla_{\boldsymbol{w}} L_i = \boldsymbol{w}, \qquad \frac{\partial L_i}{\partial b} = 0.$$

One gradient-descent step with learning rate η is

$$\boxed{\boldsymbol{w} \leftarrow (1-\eta)\boldsymbol{w}, \qquad b \leftarrow b.}$$

3.3 Case 2: Violation Gradient and Update Rule

If

$$y_i\left(\boldsymbol{w}^T\boldsymbol{x}_i + b\right) < 1,$$

the hinge term is active:

$$L_i = \frac{1}{2}\|\boldsymbol{w}\|^2 + C\left[1 - y_i\left(\boldsymbol{w}^T\boldsymbol{x}_i + b\right)\right].$$

Hence,

$$\nabla_{\boldsymbol{w}} L_i = \boldsymbol{w} - C y_i \boldsymbol{x}_i, \qquad \frac{\partial L_i}{\partial b} = -C y_i.$$

The update becomes

$$\boxed{\begin{aligned} \boldsymbol{w} &\leftarrow (1-\eta)\boldsymbol{w} + \eta C y_i \boldsymbol{x}_i, \\ b &\leftarrow b + \eta C y_i. \end{aligned}}$$

4.1 Mini-Batch Objective and Violation Set

In lab 1, we use mini-batch gradient descent to update $\boldsymbol{w}$ and b only after computing the **average gradient** over a small batch of samples, rather than after every single sample.

For a mini-batch $\mathcal{B}$ of size $B = |\mathcal{B}|$, use the mean hinge loss:

$$\boxed{J_{\mathcal{B}}(\boldsymbol{w}, b) = \frac{1}{2}\|\boldsymbol{w}\|^2 + \frac{C}{B}\sum_{i\in\mathcal{B}} \max\Big\{0, 1 - y_i\big(\boldsymbol{w}^T\boldsymbol{x}_i + b\big)\Big\}.}$$

Define the violated samples as

$$\boxed{\mathcal{V} = \Big\{i \in \mathcal{B} : y_i\big(\boldsymbol{w}^T\boldsymbol{x}_i + b\big) < 1\Big\}.}$$

Only samples in $\mathcal{V}$ contribute a hinge-loss correction. However, the regularization gradient $\boldsymbol{w}$ is still present for the batch.

Convention: the lab code uses this mean-loss form. It matches the earlier sum-loss form after a corresponding rescaling of C.

4.2 Weight and Bias Gradient for Mini-Batch

Take the mean of the 2 cases discussed in Section 3 for each sample:

$$\begin{aligned}
\nabla_{\boldsymbol{w}} J_{\mathcal{B}} &= \frac{1}{B}\left[\sum_{i\in\mathcal{B}\setminus\mathcal{V}} \boldsymbol{w} + \sum_{i\in\mathcal{V}} \left(\boldsymbol{w} - C y_i \boldsymbol{x}_i\right)\right] \\
&= \frac{1}{B}\left[\sum_{i\in\mathcal{B}} \boldsymbol{w} - C \sum_{i\in\mathcal{V}} y_i \boldsymbol{x}_i\right] \\
&= \frac{1}{B}\left[B\boldsymbol{w} - C \sum_{i\in\mathcal{V}} y_i \boldsymbol{x}_i\right] \\
&= \boxed{\boldsymbol{w} - \frac{C}{B} \sum_{i\in\mathcal{V}} y_i \boldsymbol{x}_i}.
\end{aligned}$$

$$\begin{aligned}
\frac{\partial J_{\mathcal{B}}}{\partial b} &= \frac{1}{B}\left[\sum_{i\in\mathcal{B}\setminus\mathcal{V}} 0 + \sum_{i\in\mathcal{V}} (-C y_i)\right] \\
&= \boxed{-\frac{C}{B} \sum_{i\in\mathcal{V}} y_i}.
\end{aligned}$$

There is no bias gradient for non-violating samples.

First term comes from *all* B samples; the second term is restricted to violations.

4.3 Mini-Batch Gradient-Descent Updates

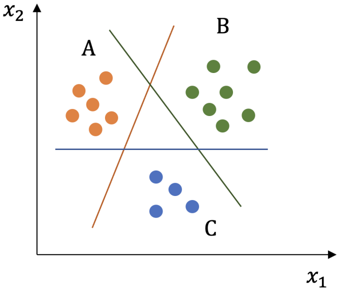
Using

$$\boldsymbol{w} \leftarrow \boldsymbol{w} - \eta \nabla_{\boldsymbol{w}} J_{\mathcal{B}}, \qquad b \leftarrow b - \eta \frac{\partial J_{\mathcal{B}}}{\partial b},$$

we obtain

$$\boxed{\begin{aligned} \boldsymbol{w} &\leftarrow (1-\eta)\boldsymbol{w} + \eta \frac{C}{B} \sum_{i \in \mathcal{V}} y_i \boldsymbol{x}_i, \\ b &\leftarrow b + \eta \frac{C}{B} \sum_{i \in \mathcal{V}} y_i. \end{aligned}}$$

Multi-Class SVM: One-vs-Rest (OvR)



A multi-class problem can be decomposed into several binary classification problems. The strategy determines which samples each binary classifier sees.

One-vs-Rest/One-vs-All (OvR/OvA)

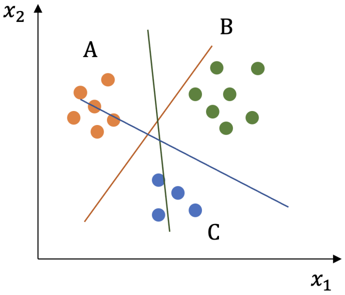
- For K classes, train K **binary classifiers**.
- Classifier k uses **all** training samples:

$$y_i^{(k)} = \begin{cases} +1, & y_i = k, \\ -1, & y_i \neq k. \end{cases}$$

- Predict using the classifier with the **largest** decision score:

$$\widehat{y} = \arg\max_k f_k(\boldsymbol{x}).$$

Multi-Class SVM: One-vs-One (OvO)



One-vs-One (OvO)

- Train one classifier for **every pair/combination** of classes:

$$\text{Total classifiers} = \binom{K}{2} = \frac{K(K-1)}{2}.$$

- Classifier (r, s) uses **only** samples from classes r and s:
  - One class is labelled $+1$.
  - The other is labelled -1.

- **Prediction:** Each pairwise classifier casts one vote. The class receiving the most votes is selected (**Majority Voting**).

Now Start the Lab

Let's get started!

(Please make sure you have scanned the attendance QR code)